

EMMANUEL MADUABUM

Electrical & Electronic Engineer · Energy Systems, Financial Modelling & Automation

+86 189 6806 4285 · +233 54 53 9998 · emmanuelmaduabum@gmail.com

Hangzhou, Zhejiang, China · [linkedin.com/in/emmanuel-maduabum](https://www.linkedin.com/in/emmanuel-maduabum) · github.com/Maanuell · maanuell.github.io

PROFILE

Electrical and electronic engineer who works on both sides of a project, the technical design and the numbers that justify it. Four years across renewable energy delivery, financial modelling and commercial operations across Ghana and Nigeria and a master's in intelligent manufacturing and control engineering in China. I design and audit energy systems, build the financial models and technical documentation that decisions get made on, and an early and deliberate adopter of AI tooling to automate reporting, data preparation and documentation which is what makes this range possible for one person.

KEY COMPETENCIES

Advanced Microsoft Excel	Financial Modelling	Data Analysis
Energy Auditing	Renewable Energy Modelling	Technical Documentation
Python & MATLAB	PVsyst & AutoCAD 2D/3D	Project Coordination

PROFESSIONAL EXPERIENCE

Business Operations Manager

Oct 2023 – Present

Cotto Enterprise · Abia State, Nigeria · petroleum lubricants trading

Runs the commercial engine of a multi-region distribution business: sales, logistics, budgeting, and compliance. Owns the numbers end to end, from the delivery schedule to the management report.

- Coordinates multi-region distribution and organization, negotiating between suppliers, transporters, and customers to keep deliveries on schedule.
- Manages budgeting, financial administration and regulatory-compliance documentation.
- Resolves scheduling and delivery constraints under time pressure, protecting customer commitments.
- Maintains and reconciles sales and inventory data across regions in Excel, turning inconsistent regional records into reporting management can act on.

Project Engineer

Mar 2023 – Dec 2023

Mcjones Energy · Accra, Ghana

Took renewable energy projects from first site visit to signed-off business case; the audit, the system design, and the financial model that decided whether the project went ahead.

- Built financial models with sensitivity analysis and income projections that underpinned client investment decisions.
- Performed energy audits identifying consumption patterns and quantifying achievable cost and energy savings.
- Designed and simulated renewable energy systems in PVsyst; produced equipment specifications and feasibility studies.
- Authored client-facing technical and financial documentation in Excel and Word for multi-party projects.
- Planned, scheduled, and monitored assigned projects against budget, timeline, and quality standards.

Junior Research Analyst

Sep 2022 – Feb 2023

HerrnHut Corp · Tema, Ghana

Provided the research and financial analysis behind a multi-site eco-center development, turning scattered source data into figures the project team could plan against.

- Contributed to successive versions of the development's consolidated financial model, including pre-tax and after-tax scenario variants.
- Produced the site-by-site power requirement analysis used to size the development's energy provision.
- Collected and cleaned research data from multiple sources into consistent datasets for analysis.
- Tracked and forecast project progress, reporting metrics and emerging risks to management.

Electrical Engineering Intern

Sep 2021 – Feb 2022

GFEEL Engineering · Tema, Ghana

Supported design, installation, and maintenance of electrical systems across residential, commercial, and industrial sites — wiring, CCTV, solar and electrical fencing — contributing technical drawings, cost estimates, and field inspections.

PROJECTS

Autonomous Lawn Mower Robot *self-built platform · hardware complete, autonomy in development*

- Designed, built, and wired a working outdoor robot from scratch, chassis, drive system, power, and control electronics.
- Building the navigation stack on ROS with RTAB-Map for mapping and localization and evaluating reinforcement learning for coverage path planning.
- Solving the failure modes that only appear outdoors: uneven terrain, glare, and how completely a site changes between seasons.

Green Hydrogen & Fuel Cell Feasibility Study *technical and economic study*

- Assesses coupling an electrolyser and fuel cell to surplus renewable and waste-to-energy generation to store and firm intermittent power.
- Covers technology selection, indicative sizing against available surplus power and water, and the point at which a demonstration unit becomes economically defensible.

Solar Irradiance Forecasting with Deep Learning *MATLAB · LSTM, ConvLSTM · time-series modelling.*

- Built and benchmarked recurrent and convolutional-recurrent models for short-horizon irradiance prediction.

Cloud Classification with Convolutional Neural Networks *MATLAB · CNN · image processing*

- Trained a CNN to classify sky-image cloud types — the perception half of a solar nowcasting system.

Sales Performance Dashboard *Microsoft Power BI · Dataleum team project*

- Turned a raw transactional dataset into an interactive dashboard: cleaning and shaping, calculated measures, and visuals built for a non-technical audience.

Speed-Bump Energy Generator for Street Lighting *mechanical design · energy conversion · system modelling.*

- Designed a road-mounted harvester converting vehicle motion into electricity for street lighting, including the conversion mechanism and system-level modelling.

EDUCATION

Master of Engineering (MEng), Intelligent Manufacturing and Control Engineering

Sep 2025 – Jan 2028

Zhejiang University of Science and Technology · Hangzhou, Zhejiang, China

- Awarded the Scholarship for Outstanding New Students (50% of first-year tuition).
- Research area: probabilistic solar irradiance forecasting and autonomous systems.
- Minister of Sports & Arts Department, ZUST International Student Union.

Bachelor of Engineering (BEng), Electrical and Electronic Engineering

2019 – 2023

Accra Institute of Technology · Accra, Ghana

CERTIFICATES

- Data Analytics and Business Intelligence — Dataleum (January 2024)
- Project Development for Renewable Energies — Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH (January 2023)
- Duolingo English Test: 120, CEFR B2 (December 2024)

LANGUAGES

- English (fluent) · Igbo (native) · Mandarin Chinese (HSK 2, improving)

REFERENCE

Mr. Jones Elias Ntorinkansah

Managing Director & CEO, Mcjones Energy, Accra, Ghana

+233 (0) 55 252 3087 · +44 (0) 7494 65 77 45